

and liquidity buffers. As benchmarks, safe assets support the pricing of other riskier assets. Finally, safe assets have been a critical component of monetary policy operations.

Whew! Let's just discuss a few factors:

Credit Risk

An ideal store of value has zero default risk. The private sector, then, has great difficulty creating enough safe assets (the literature calls these *information-insensitive assets*) to meet demand.³⁹ But sovereign debt also has credit risk. Sovereigns default all the time.⁴⁰ Tomz and Wright (2013) compute that there is a 1 in 50 chance that a given country is in default at any time, but for the post-1980 sample the annual probability of default more than doubles, to 4%. Investors lose around 40% (the *haircut*) when sovereigns default, but there is wide variation with a range of 30% to 75%.⁴¹

Even the United States has defaulted. Twice. In 1934, it slashed the value of a dollar from \$20.67 per troy ounce of gold to \$35. Reinhart and Rogoff (2008) classify the U.S. abrogation of the gold clause as a default, which it was because at the time all major currencies were backed by gold. Then, in a little-known episode in April and May 1979, T-bill investors did not receive their interest payments on time.⁴² The Treasury blamed processing problems and the failure of Congress to act on the debt ceiling (the latter is a perennial problem). At first, the Treasury did not want to pay any additional interest, which triggered a class-action lawsuit. The government paid only after much lobbying.

If we were to rank countries on credit risk, then we might rank three (I've randomly selected these) countries in order of most safe to least safe as: Norway, United States, and Brazil.

Collateral

Safe assets are used as collateral to borrow money or short other assets.⁴³ Ranking the debts of countries on the ability to use them as collateral we have: the United States (Treasury bills are by far the largest source of such collateral), followed by Norway and Brazil depending on specific circumstances (like whether you're a Norwegian or Brazilian investor).

³⁹ See Gorton (2010).

⁴⁰ Reinhart and Rogoff (2011) is a comprehensive treatise. They show that from the 1930s to the 1950s, nearly half of countries were in default.

⁴¹ See Sturzenegger and Zettelmeyer (2008) and Cruces and Trebesch (2013).

⁴² Zivney and Marcus (1989) examine this interesting episode.

⁴³ A model of the collateral role for safe assets is Gorton and Ordonez (2013).